

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India **09/2021 – 01/2025 (3.5 Years)**

- Developed and implemented **System Verilog Assertions (SVA)** for complex digital design blocks on AI accelerators, achieving **98% assertion coverage** on critical logic paths.
- Engineered **System Verilog** verification environments from planning through signoff, reducing verification cycle time by **20%** through modular testbench design.
- Created **random test generators** and **high-level transactional models** in System Verilog, increasing bug discovery by **15%** compared to directed testing alone.
- Integrated **Python scripting** into CI/CD pipelines (Jenkins) for automated test execution and result aggregation, decreasing manual intervention by **40%**.
- Managed change-control processes for enterprise business systems and ERP modules, coordinating UAT and post-go-live support across interconnected platforms.
- Analyzed and resolved production incidents for financial-services dashboards, performing log analysis in **Linux** environments and utilizing **Splunk** for root cause identification, meeting **99.9% SLA** targets.

SKILLS

Verification Languages – System Verilog, System Verilog Assertions (SVA)

Verification Methodologies – Functional Verification, Formal Verification, Assertion-Based Verification, UVM

Programming & Scripting – Python, C++, Shell Scripting, TCL

Hardware Concepts – ASIC, SoC, FPGA, NPU AI Accelerators, NoCs, AMBA Protocols

Tools & Environments – Jenkins, Git, Jira, Linux, Docker

Data & Analytics – SQL, Power BI, DAX, Tableau

Enterprise Systems – ERP Support, REST APIs, Change Management

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin **02/2026**

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India **01/2021**

PROJECTS

AI Accelerator Pre-Silicon Verification Platform (HCL Technologies)

- Architected and built a comprehensive **System Verilog** verification environment for an AI accelerator's core processing unit, integrating **System Verilog Assertions (SVA)** to define and check design properties.
- Implemented **random test generators** and **transactional models** using Python and System Verilog, increasing test case variety and uncovering **5 critical bugs** before tape-out.

- Developed **formal verification** constraints and properties, achieving **95% property coverage** for complex control logic and reducing simulation time for specific scenarios by **30%**.

Financial Transaction Monitoring Dashboard Enhancement (HCL Technologies)

- Enhanced a payments operations dashboard by developing new Python-based ETL processes to ingest and normalize transaction data from disparate sources, improving data availability by **25%**.
- Integrated **Splunk** alerts for critical transaction failures and reconciliation discrepancies, reducing average incident response time by **15%** and ensuring adherence to strict operational SLAs.
- Refactored SQL queries across PostgreSQL and Oracle databases to optimize report generation for reconciliation workflows, cutting report execution times by an average of **20%**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)